

Surveying

BRANCH: CIVIL ENGG. (CE)

COURSE CODE: CE202

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Surveying (CE202) under the Civil Engg. (CE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Introduction & Linear Surveying

Principles of surveying. Chain surveying: equipment, ranging, tape corrections. Compass surveying: local attraction, magnetic declination, bearing conversions.

Unit 2: Levelling & Contouring

Definitions, Levelling instruments, methods: Rise and Fall, Height of Instrument. Contour characteristics, methods of contouring, contour maps interpretation.

Unit 3: Theodolite & Curves

Theodolite traversing: horizontal, vertical angles, Bowditch rule adjustments. Curves: Simple circular curves, setting out methods (Rankine's method). Transition and Vertical curves.

Unit 4: Modern Surveying Methods

Tacheometric surveying. Total Station: components, distance and angle measurements. GPS and GIS introduction, photogrammetry basics.

*True length = Measured length * (L' / L). HI = BM + BS ; RL = HI - FS. Curvature correction = $0.0785 * d^2$.*

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers

